AI in HealthCare

CSET 343

📋 Problem Statement

Hypertension (high blood pressure) is a leading global risk factor for cardiovascular disease, heart attacks, and strokes, affecting over 1.2 billion people worldwide. It is frequently referred to as a "silent killer" because most individuals show no symptoms until severe, irreversible organ damage has already occurred. Current healthcare approaches rely on passive blood pressure recording after hypertension has already developed, while traditional online health calculators offer static, unexplainable risk scores without actionable lifestyle guidance. Furthermore, existing dietary advice is generic and fails to demonstrate to patients how specific daily habits directly impact their blood pressure. **HyperShield AI** is an intelligent, web-based healthcare platform that predicts early hypertension risk stages (Normal, Pre-Hypertension, Stage 1, Stage 2) using machine learning on lifestyle, dietary, and vital parameters. It integrates SHAP-based explainable AI feature attribution, interactive "what-if" sodium reduction simulation, and automated risk-adaptive DASH diet meal planning on one unified platform.

✅ Problems We Resolve

#	Problem Resolution
1	Late hypertension detection - Early pre-hypertension risk screening using lifestyle & vitals
2	Passive manual BP logging - Predictive machine learning classification (Normal vs Stage 1 & 2)
3	Unexplainable "black box" AI scores - SHAP visual charts showing exact risk factor breakdown
4	Unrealistic lifestyle advice - Interactive "What-If" sodium & exercise reduction simulator
5	Generic static diet charts - Risk-adaptive automated DASH-diet meal plan generator
6	Lack of patient tracking history - Longitudinal risk score tracking & health history dashboard
7	Isolated patient data - Automated 1-click PDF clinical summary report for doctor consultation
8	High platform complexity - Simple, responsive web dashboard with interactive sliders

Team Details:

Amandeep Verma E23CSEU1483
Pujith Sri Sai Karri E23CSEU1410